RISING INNOVATORS RESEARCH SHOWCASE

The INTERNATIONAL CONFERENCE for HIGH PERFORMANCE COMPUTING NOV 15–20
CHICAGO, IL NETWORKING, STORAGE, & ANALYSIS

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Abstract, Portable, Performant

realizing sustainable HPC

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The Sustainability Problem

30

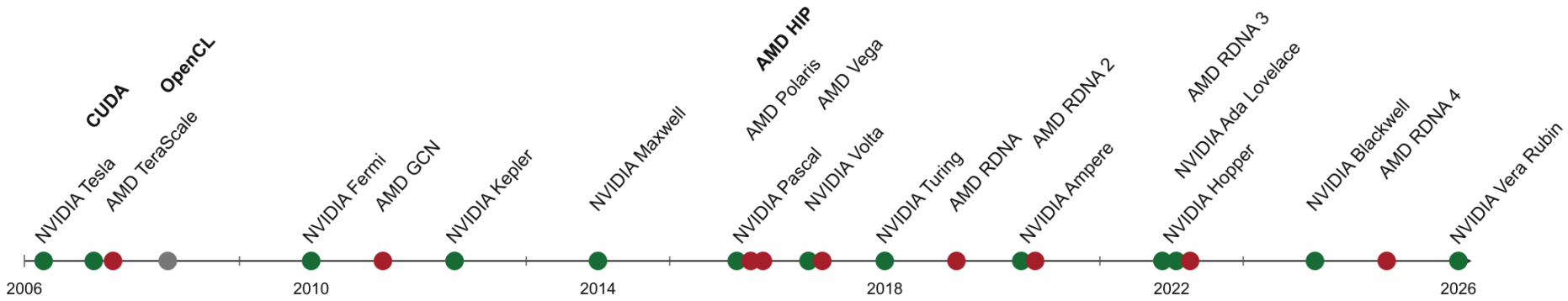
Years: average age of widely used HPC applications

5

years until a leading supercomputer is succeeded

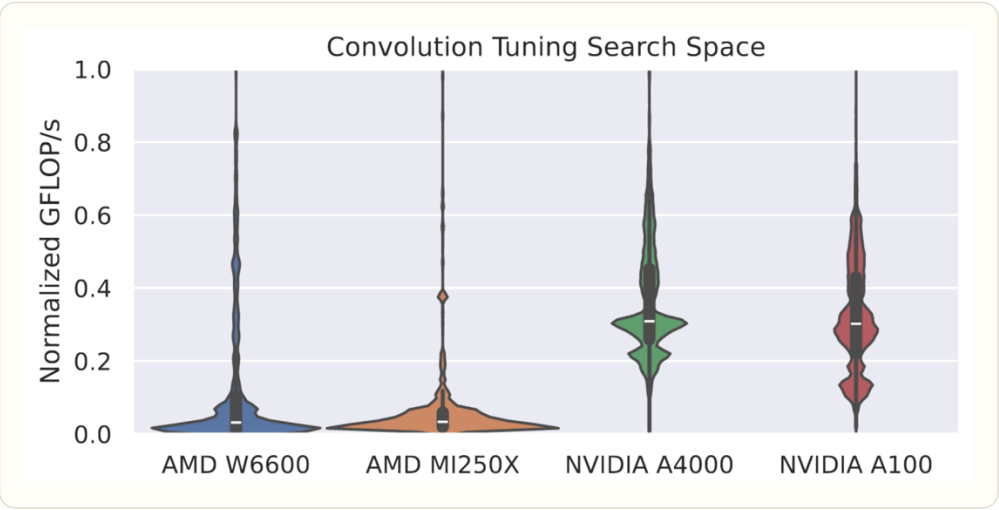
2

years between new GPU micro-architectures



Applications: VASP, LAMMPS, cp2k, GROMACS, NEMO (UK Archer2 usage data). Systems: Jaguar to Discovery, Sequoia to El Capitan.

Autotuning Impact



~50×

between the best configuration and the search-space average

But spaces hold **millions to billions** of configurations.

Lurati et al., “Bringing Auto-Tuning to HIP: Analysis of Tuning Impact and Difficulty on AMD and Nvidia GPUs,” Euro-Par 2024

Making Autotuning Scale

SC26 hpc unites. CHICAGO

01 · Getting to the top fast

BO

Bayesian optimization adapted to GPU search spaces.

02 · Measuring performance

Fair

Comparing optimizers over real time, across search spaces.

03 · Finding the gaps

>20,000×

Faster than brute force at finding valid configurations.

04 · Avoiding the gaps

~40%

Faster convergence with constraint-aware optimizers.

05 · Tuning the Tuner

~200%

Better-than-average hyperparameters; 3 years of tuning done in ~2 days.

06 · Auto-designed algorithms

LLM

LLM-designed optimizers reach best-ever performance.

The New Paradigm

01

Abstract

Write the program once, free of hardware-specific details.

02

Portable

The same source runs on a wide variety of GPUs.

03 · now feasible

Performant

Autotuning adapts the kernel to each device and environment.

Abstraction and portability on their own is not enough. Scalable autotuning is the missing piece of the puzzle.

In Practice


Abstract
Written once
Your Program
Hardware-agnostic Julia
with KernelAbstractions.jl
Portable
Compiled per vendor
CUDA · NVIDIA
ROCm · AMD
oneAPI · Intel
Metal · Apple
Performant
Tuned per device with KernelTuner.jl
A100
B200
RTX 5090
MI250X
MI300X
W7800
Max 1550
Arc A770
Arc B580
M1
M3 Pro
M5 Max

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Join the Julia for HPC tutorial tomorrow 8:30am - 5:00pm or visit:



https://github.com/KernelTuner/KernelTuner.jl